Supplementary Appendix A - Search Results

PubMed – 3.17.2021		
Search #	Query	Results
#1	(Co-Design[tiab] OR Co-Creation[tiab] OR Co-Production[tiab] OR Participatory Design[tiab] OR Human Factors[tiab] OR User-Centered Design[mesh] OR User-Centered Design[tiab] OR Participatory Approach[tiab] OR Human-Centered Design[tiab] OR Universal design[mesh] OR Patient Participation[mesh] OR patient participation[tiab])	38,538
#2	(Aged[mesh] OR Older Adults[tiab] OR Older People[tiab] OR Elderly[tiab] OR Geriatric[tiab] OR Aged over 60[tiab] OR aged 60[tiab] OR Seniors[tiab] OR Over 65[tiab])	3,333,143
#3	(Electronic Health Records[mesh] OR Electronic Health Records[tiab] OR Electronic Health Record[tiab] OR Electronic Healthcare[tiab] OR eHealth[tiab] OR "mHealth"[tiab] OR Decision Aid[tiab] OR decision aids[tiab] OR Electronic Healthcare Aid[tiab] OR Healthcare Application[tiab] OR digital health[tiab] OR health information technology[tiab] OR health information technologies[tiab] OR health technology[tiab] OR health technologies[tiab])	59,271
#4	#1 AND #2 AND #3	495

Embase – 3.17.2021		
Search #	Query	Results
#1	('human factors research'/exp OR 'user-centered design'/exp OR 'Universal design'/exp OR 'Patient Participation'/exp OR ('co-design' OR 'Co-Creation' OR 'Co-Production' OR 'Participatory Design' OR 'Human Factors' OR 'user-centered design' OR 'Participatory Approach' OR 'Human-Centered Design' OR 'patient participation'):ab,ti)	42,464
#2	('aged'/exp OR ('older adult' OR 'older adults' OR 'older people' OR elderly OR geriatric OR geriatrics OR 'aged over 60' OR 'aged 60' OR 'over 60' OR 'aged over 65' OR 'aged 65' OR 'over 65' OR senior OR seniors):ab,ti)	3,460,578
#3	('electronic health record'/exp OR ('electronic health record' OR 'electronic health records' OR 'electronic healthcare' OR 'ehealth' OR 'mhealth' OR 'electronic decision aid' OR 'electronic decision aids' OR 'electronic healthcare aid' OR 'electronic healthcare aids' OR 'healthcare applications' OR 'Decision Aid' OR 'decision aids' OR 'digital health' OR 'health information technology' OR 'health information technologies' OR 'health technology' OR 'health technologies'):ab,ti)	59,924
#4	#1 AND #2 AND #3	246

Scopus – 3.17.2021		
Search #	Query	Results
#1	TITLE-ABS("co-design" OR "Co-Creation" OR "Co-Production" OR "Participatory Design" OR "Human Factors" OR "user-centered design" OR "Participatory Approach" OR "Human-Centered Design" OR "patient participation")	78,052
#2	TITLE-ABS("older adult" OR "older adults" OR "older people" OR elderly OR geriatric OR geriatrics OR "aged over 60" OR "aged 60" OR "over 60" OR "aged over 65" OR "aged 65" OR "over 65" OR senior OR seniors)	657,056

#3	TITLE-ABS("electronic health record" OR "electronic health records" OR "electronic healthcare" OR "ehealth" OR "mhealth" OR "electronic decision aid" OR "electronic decision aids" OR "electronic healthcare aid" OR "electronic healthcare aids" OR "healthcare applications" OR "Decision Aid" OR "decision aids" OR "digital health" OR "health information technology" OR "health information technologies" OR "health technology" OR "health technologies")	54,356
#4	#1 AND #2 AND #3	94

- records before duplicates removed: 835
- records after duplicates removed: 625

Supplementary Appendix B - Study Classification Methods

Category	*Variable*	*Classification method*
Summary characteristics	Year of publication	Metadata
	First author's geolocation	Metadata
	Study setting	Metadata
	Number of stakeholders	Metadata, mean
	Number of older adults	Metadata, mean
	Study population	Metadata
Co-design approaches	Terminology or definitions	Manual, binary
	Theoretical framework/design principles	Manual, binary
	Iterative development	Metadata, mean
Stakeholder involvement	Participant types	Metadata
	Types of Involvement Processes	Leinonen et.al [24]
	Learning Levels	Bateson [25], Argyris [26]
	Levels of participation	Vaughn et.al [3]
Electronic healthcare tools	Types of healthcare tools	WHO classification of digital health interventions [17]
	User testing	Metadata

Supplementary Appendix C

Author	Theoretical Framework/Co-Design Principle	Level of Involvement	Involvement Processes																User Testing Measures and Validation					
			A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	Endpoint	Other	Objective Measure	Subjective Measure	Physiological	Validated
Abujarad [52]	User Centered Design Approach	Empower	X	X	X	X	X			X					X	X		X		confidence and computer efficacy			CES		X
																				emotional reactions			I-PANAS-SF		X
																				usability			SUS		X
Ahmed [34]	Participatory Design Approach Developed recommendations for framework	Empower	X	X					X			X	X			X		X		health literacy			NVS		X
																				patient engagement - self report on health behaviors			ACE (Altarum Consumer Engagement)		X
Bogza [29]	Center for eHealth and Wellbeing Research guidelines Inductive thematic qualitative analysis approach IPDAS	Empower			X	X	X								X	X		X	X	quality of decision aid			Ottawa Acceptability Scale		X
																				satisfaction			Likart - unspecified		
																				usability			SUS		X
																							Decisional Conflict Scale		X
																				Comprehension	teach back method				
Cornet [12]	User Centered Design Developed recommendations for framework	Involve		X	X	X	X			X	X			X		X	X			cognitive workload			NASA-TLX		X
																				health literacy			NVS		X
																				usability			SUS		X
																				user acceptance			Likart - unspecified		
																				errors		Not stated			
Gonzalez [33]	none	Involve			X											X				appearance			Likart - unspecified		
																				comfort			Likart - unspecified		
																				functionality			Likart - unspecified		
																				reliability			Likart - unspecified		
																				usability			Likart - unspecified		
Grimaldi [11]	Human computer design approach ISO 13407:1999 [1] revised by ISO 9241-210:2010	Collaborate		X	X	X	X			X				X		X		X		quality of interaction			Us.E 2.0		
																				satisfaction			Likart - unspecified		
																				usability			Likart - unspecified		
																				errors		Not stated			
Gustafson [14]	NIATx model as a framework Developed recommendations for framework	Empower	X		X	X			X			X			X	X		X	X	usability	Spending time teaching older adults how to use technology in a classroom		Likart - unspecified		

																			setting					
Harrington [9]	Experience-based co-design Developed recommendations for framework	Empower	X	X	X				X				X					X		usability			Likart - unspecified		
																			usefulness	Likart - unspecified				
Harte [7]	HCD methodology ISO 9241-210 standards	Empower	X	X	X	X				X	X	X		X		X	X	X	cognitive workload			NASA-TLX		X
																			efficiency			ASQ		X
																			satisfaction			Likart - unspecified		
																			usability			SUS		X
																			errors		Mistakes Counted			
																			Performance		Time to Complete			
Hoffman [39]	IPDAS Ottawa Decision Support	Empower		X	X	X	X		X		X	X			X	X	X	X	quality of decision aid			Ottawa Acceptability Scale		X
																			values clarity			SURE		X
																			errors		Observed errors			
Holden [53]	Age-Related Research-Based Usability Guidelines Principles and Creative Human Factors Approaches	Empower		X	X		X			X					X	X		X	usability			SUS		X
																			errors		Mistakes Counted			
																			need for assistance		Times Request Assistance			
																			Performance		Time to Complete			
Kim [38]	Grounded theory approach	Involve	X	X	X				X			X	X		X									
Mansson [35]	Karagianni's Optimized Honeycomb model	Empower					X			X			X			X		X	findable, accessible, usable, desirable, credible, useful, and valuable	UX Honeycomb Model ALL QUALITATIVE				
Martin-Hammond [36]	Nielsen's Heuristics	Involve			X	X					X	X		X		X	X		usability	qualitative				
Nguyen [42]	Medical Research Council framework Developed recommendations for framework	Empower			X		X			X			X		X	X		X	satisfaction			Likart - unspecified		
																			errors		Observed errors			
																			Navigation	Google Analytics				
Nielsen [44]	HCD International Organization for Standardization (ISO) standards for human-centered design for	Empower							X			X			X	X		X	usability			Likart - unspecified		

	interactive systems (European Committee for Standardization, 2010) and applied principles and techniques from the user innovation management method (Kanstrup & Bertelsen, 2011).																	user acceptance			Likart - unspecified		
Or [43]	Human Factors Design principles Nielson heuristic principles Systematic usability study guideline	Collaborate			x		x			x						x	x			ease of use			Likart - unspecified		
																				satisfaction			Likart - unspecified		
																				usefulness			Likart - unspecified		
																				errors		Mistakes Counted			
																				Need for assistance		Times Request Assistance			
																				Performance		Time to Complete			
Petersen [45]	IPDAS	Empower			x		x									x		x		comprehensibility			Likart - unspecified		
																				effectiveness			Likart - unspecified		
																x				satisfaction			Likart - unspecified		
																				usability			Likart - unspecified		
																				value of the information			Likart - unspecified		
Poirier [28]	IPDAS	Consult																		comprehension /recall			Likart - specified validity		X
																				gaze fixation				eye-tracking	
										x	x								x	graphicacy			Likart - specified validity		X
																				numeracy			Likart - specified validity		X
																				values clarity			Decisional Conflict Scale		X
																				performance		Time to Complete			
Portz [40]	Grounded theory approach Human-Centered Design framework	Involve			x		x																		
Scandurra [37]	Human-Computer Interaction (HCI) research theory	Empower											x					x		share experience with others			Likart - unspecified		
																				usability			Likart - unspecified		
																				user acceptance			Likart - unspecified		
Ummels [41]	Double Diamond Method	Empower			x			x	x	x						x		x		satisfaction			Likart - unspecified		

	MoSCoW model Ummels Framework Developed recommendations for framework																	usability			Likart - unspecified		
Van De Dijk [54]	User Centered Design Approach	Involve		x				x		x		x					x						
Van Velsen [55]	MoSCoW method	Involve								x	x	x		x				usability	qualitative				
Wikberg-Nilsson [8]	None	Empower			x							x			x		x	cognitive process				eye-tracking	
																		usability	qualitative				

A. Contextual Inquiry
B. Survey
C. Interview
D. Cognitive walkthrough
E. Think aloud
F. Remote evaluation
G. Focus group
H. Task analysis
I. Scenarios
J. Prototype (low fidelity)
K. Working Group Workshop
L. Use Cases
M. User Stories
N. Prototype (medium fidelity)
O. Heuristic evaluation
P. Prototype (high fidelity – functional/Beta version)
Q. Other Process

Supplementary Appendix D

Co-Design Approach	***Inform***		***Consult***		***Involve***		***Collaborate***		***Empower***	
	n	ref	n	ref	n	ref	n	ref	n	ref
Co-creation (n=2)	**2**	[35,41]	**2**	[35,41]	**2**	[35,41]	**2**	[35,41]	**2**	[35,41]
Co-design (n=5)	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]	**5**	[8,9,39,42,44]
Human centered computing (n=1)	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]
Human Centered Design (HCD) (n=3)	**3**	[7,11,40]	**3**	[7,11,40]	**3**	[7,11,40]	**2**	[7,11]	**1**	[7]
Human Factors (n=1)	**1**	[33]	**1**	[33]	**1**	[33]	**0**		**0**	
Interaction Design Process (n=1)	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]	**1**	[9]
Participatory Design (n=7)	**7**	[8,9,34,37,38,44,55]	**7**	[8,9,34,37,38,44,55]	**7**	[8,9,34,37,38,44,55]	**5**	[8,9,34,37,44]	**5**	[8,9,34,37,44]
Usability Evaluation Approach (n=1)	**1**	[43]	**1**	[43]	**1**	[43]	**1**	[43]	**0**	
User Centered Design/Approach (n=9)	**9**	[12,14,29,36,39,41,52–54]	**9**	[12,14,29,36,39,41, 52–54]	**9**	[12,14,29,36,39,41, 52–54]	**5**	[14,29,39,41,53]	**6**	[14,29,39,41,52,53]
Other (Did not Specify) (n=2)	**2**	[28,45]	**2**	[28,45]	**1**	[45]	**1**	[45]	**1**	[45]